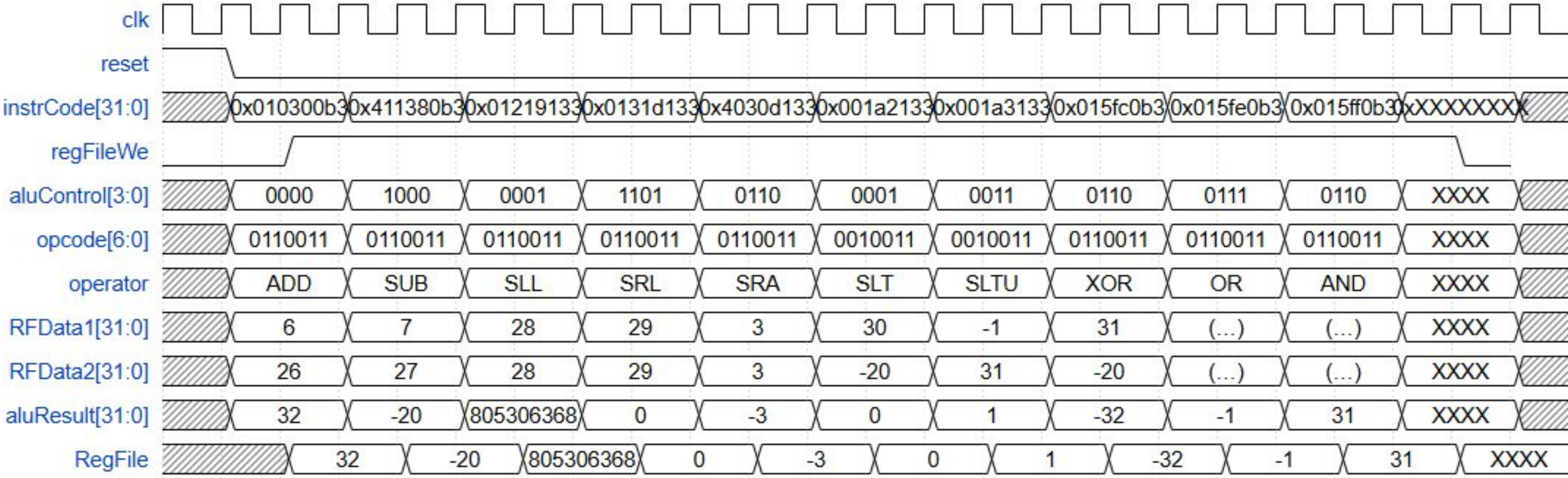
Timing Diagram



<초깃값 정리>

mem[0:15]: 0~15

mem[16:30]: 10 + 16~30

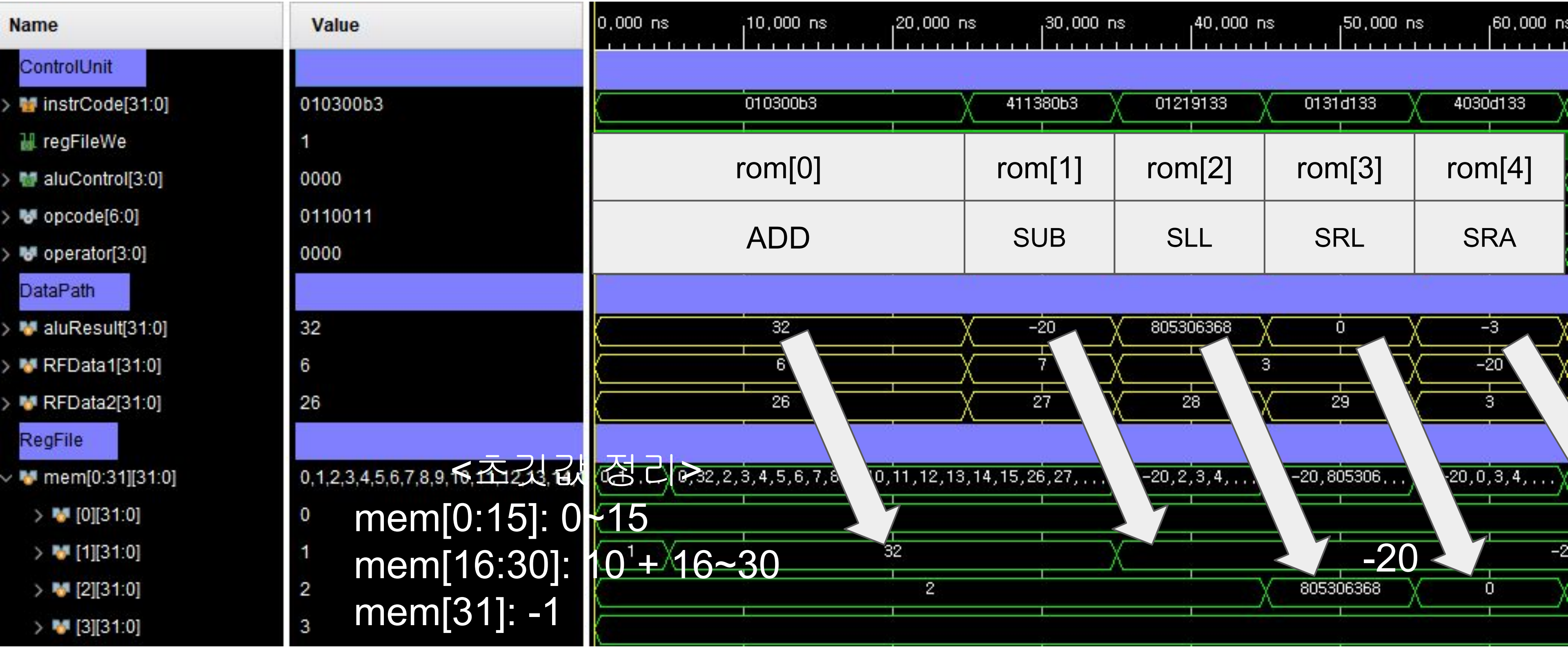
mem[31]: -1

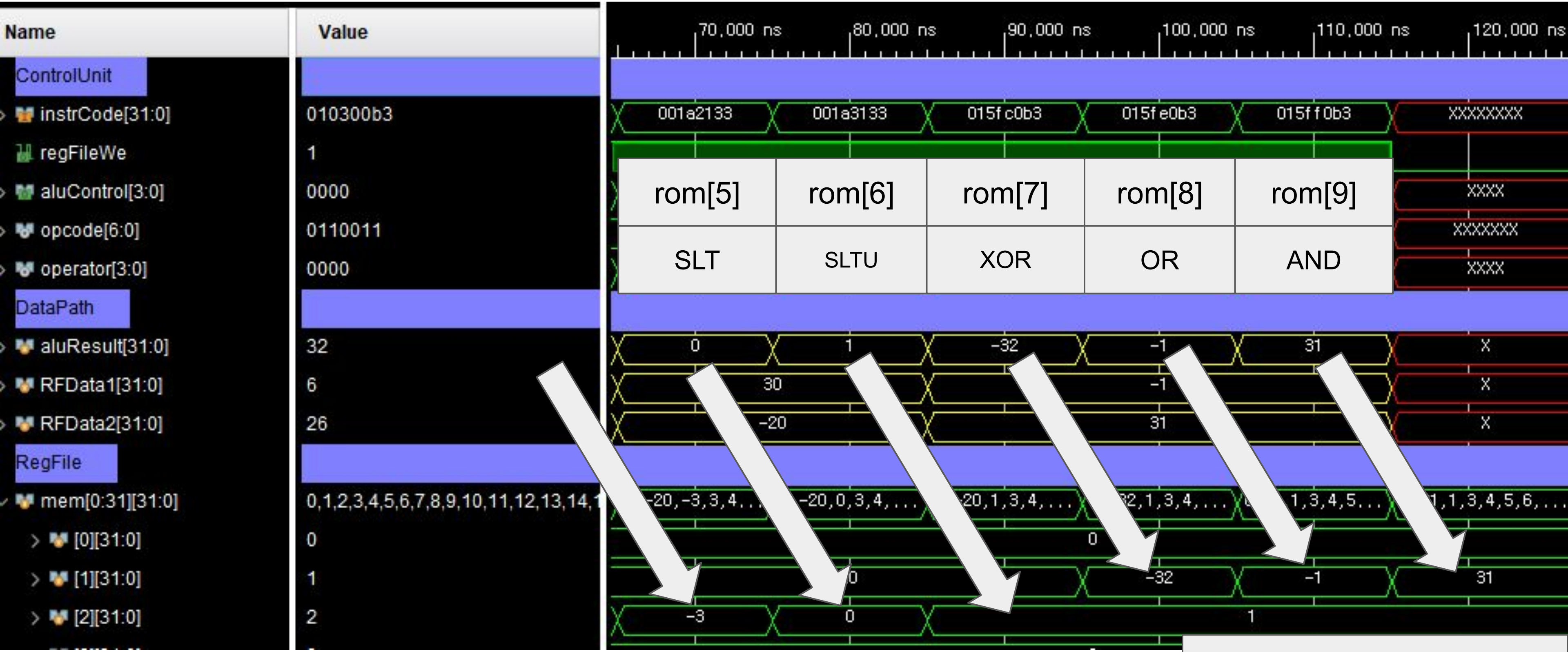
```
initial begin
    //rom[x]=32'b func7 _ rs2 _ rs1 _fu3_ rd  _ op
    rom[0] = 32'b0000000_10000_00110_000_00001_0110011;//add x16, x6, x1
    rom[1] = 32'b0100000_10001_00111_000_00001_0110011;//sub 17, x7, x1
    rom[2] = 32'b0000000_10010_00011_001_00010_0110011;//sll x18, x3, x2
    rom[3] = 32'b0000000_10011_00011_101_00010_0110011;//srl x19, x3, x2
    rom[4] = 32'b0100000_00011_00001_101_00010_0110011;//sra x3, x1, x2
    rom[5] = 32'b0000000_00001_10100_010_00010_0110011;//slt x1, x20, x2
    rom[6] = 32'b0000000_00001_10100_011_00010_0110011;//sltu x1, x20, x2
    rom[7] = 32'b0000000_10101_11111_100_00001_0110011;//xor x21, x31, x1
    rom[8] = 32'b0000000_10101_11111_110_00001_0110011;//or x21, x31, x1
    rom[9] = 32'b0000000_10101_11111_111_00001_0110011;//and x21, x31, x1
end

assign data = rom[addr[31:2]];
endmodule
```

시뮬레이션 전,
예측값

rom	op	rs1	rs2	rd
0	ADD	x6(6)	x16(26)	x1(1->32)
1	SUB	x7(7)	x17(27)	x1(32->-20)
2	SLL	x3(3)	x18(28)	x2(2->805306368)
3	SRL	x3(3)	x19(29)	x2(805306368->0)
4	SRA	x1(-20)	x3(3)	x2(0->-3)
5	SLT	x20(30)	x1(-20)	x2(-3->0)
6	SLTU	x20(30)	x1(-20)	x2(0->1)
7	XOR	x31(-1)	x21(31)	x1(-20->-32)
8	OR	x31(-1)	x21(31)	x1(-32->-1)
9	AND	x31(-1)	x21(31)	x1(-1->31)


Name
Value
ControlUnit
instrCode[31:0]
010300b3
regFileWe
1
aluControl[3:0]
0000
opcode[6:0]
0110011
operator[3:0]
0000
DataPath
aluResult[31:0]
32
RFData1[31:0]
6
RFData2[31:0]
26
RegFile
mem[0:31][31:0]
[0][31:0]
0
[1][31:0]
1
[2][31:0]
2
[3][31:0]
3
0,000 ns
10,000 ns
20,000 ns
30,000 ns
40,000 ns
50,000 ns
010300b3
411380b3
01219133
0131d133
4030d133
rom[0]
rom[1]
rom[2]
rom[3]
rom[4]
ADD
SUB
SLL
SRL
SRA
32
-20
805306368
0
-3
6
7
3
-20
26
27
28
29
3
32
2
805306368
0
-20
<초깃값 정리>
mem[0:15]: 0~15
mem[16:30]: 10 + 16~30
mem[31]: -1


Name
Value
70,000 ns
80,000 ns
90,000 ns
100,000 ns
110,000 ns
120,000 ns
ControlUnit
instrCode[31:0]
010300b3
001a2133
001a3133
015fc0b3
015fe0b3
015ff0b3
XXXXXXXX
regFileWe
1
aluControl[3:0]
0000
opcode[6:0]
0110011
operator[3:0]
0000
rom[5]
rom[6]
rom[7]
rom[8]
rom[9]
SLT
SLTU
XOR
OR
AND
DataPath
aluResult[31:0]
32
0
1
-32
-1
31
RFData1[31:0]
6
30
-1
RFData2[31:0]
26
-20
31
RegFile
mem[0:31][31:0]
0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,1
[0][31:0]
0
[1][31:0]
1
-32
-1
31
[2][31:0]
2
-3
0
1

예측값과 일치